

IT Hardware Industry

Expert Call Series Takeaways: Optical in the data center

Industry Overview

Featured Mr. Frank Chang on optical in the data center

As part of our 2026 BofA Hardware Expert Call Series, we host industry experts across various topical areas. We hosted Mr. Frank Chang, CTO of Source Photonics (advanced tech solutions provider for comm & data connectivity focused on optical transceivers). Transcript from the call is in the back of this note.

AI optical demand primarily scale-out & scale-across

AI optical demand remains extremely strong, but the opportunity is still primarily concentrated in scale-out & scale-across vs. scale-up. While there is significant interest in optical scale-up, Mr. Chang's view was that major architectures, especially Nvidia's, will continue using copper as long as possible, likely through at least 2028 in mainstream deployments. Within optics, the current cycle is being led by 800G, with 1.6T emerging as the next major ramp. 3.2T is the next long-term node but still further out. As rough modeling frameworks, Mr. Chang pointed to a linkage from data center power to GPU count (~100MW equating to ~100k GPUs), from GPUs to transceiver count (~5-6x today & potentially ~8-9x in denser 1.6T architectures), and from transceiver count to spend using ASP assumptions (~\$350 for 800G & ~\$600-\$700 for 1.6T). Mr. Chang also suggested that short-reach copper remains dominant in scale-up through 2028, AEC becomes more important at 1.6T for roughly 5m to 7m reaches, and actual shipments may be capped by component availabilities while lead times stretches.

Demand highly differentiated by each hyperscale customer

Hyperscaler demand remains highly differentiated by customer. AWS was described as still heavily engaged in 400G/800G while moving somewhat more cautious than others. Google is further along in newer optical architectures and 1.6T with OCS being more central in design choices. Meta is a major volume opportunity in 2026 for Source Photonics with 800G ramps in 2026 & potential openness to newer optical approaches vs. peers. Microsoft is a large NVDA-linked buyer & uses a more distributed DC model in some cases & important in certain longer-reach or campus/inter-DC optical links. China cloud remains an optical market as well but appears to be running a few years behind the leading US customers & still centered more heavily on 400G. Mr. Chang suggested that the industry is already facing bottlenecks in lasers and related optical components, with shortages potentially lasting into 2027.

Pluggables remain dominant for now, CPO to come later

Mr. Chang was cautious on CPO & stated that pluggables will remain dominant for years because hyperscalers understand how to qualify & service them vs. CPO which presents significant complexity in validation, repairability, & supplier flexibility. Instead, the nearer-term bridge techs may be AEC in short reach & newer advanced pluggable form factors beyond today's standards (XPO was discussed as a pluggable optical form factor that could preserve serviceability & ecosystem flexibility). DAC runs out of reach / signal integrity at higher speeds & 1.6T creates a strong use case for AEC at intermediate distances, which in turn makes AEC relevant for some scale-up & short-reach scenarios before optics.

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Abbreviations included in back

Transcript

Wamsi Mohan: Thank you so much. Good afternoon, everyone. Lots happening in the market, so we appreciate you taking the time to join us today. This is another call in our series of BofA Hardware Expert Call Series. A quick disclaimer before we dive in. Conflict disclosures related to the individual companies or securities discussed on the call today can be found on the call invitation. So today, we're delighted to welcome Frank Chang. Frank is the CTO of Source Photonics since 2018. The company provides advanced technology solutions for communications and data connectivity. He's had prior roles as Senior Principal Engineer for Optica at InPhi. Frank, actually, I believe is in China right now, so he is definitely available on audio. I'm not sure all the bandwidth for him on video, but we're happy to welcome Frank today. So, Frank, thank you so much for taking the time to join us.

Frank Chang: Thank you for the introduction. I appreciate setting up this call.

Wamsi Mohan: Frank, do you want to just give 30 seconds of your background so everyone on the call can get a little bit familiar with your expertise?

Frank Chang: Sure, definitely. My name is Frank Chang. I work for Source Photonics. So, if you know Source Photonics before, Source Photonics is a company running basically the hope the optical transceiver vendor. So, I think we are kind of a top nine right now for the industry. We shape like, kind of like meanings. I think, right now probably 4 million to 5 million units, 800G, 400G transceivers for each year. So, basically, I think, since 2023 all revenue doubled every year. So, I think 2025 we're like an \$800 million company USD. So, we have three divisions, basically three locations for the industry. First, for example, we have US headquarters. We're shaping our manufacturer and the transceivers and also the fabs. We have optical chip fabs in Taiwan, and also we have China, two sites in China, Chengdu and Jin tan. So, regarding the optical transceiver, the developments, and also the foundry. So that's our general situation. I think many of you are probably familiar, OFC is coming in March. So, we will introduce our new products, especially for the 800G 1.6T. And also, we will download the single Lambda 400G products, and that will prepare for the 3.2T optical transceivers.

I'm not sure what else you want to hear.

Wamsi Mohan: That's a great introduction. Thank you. So maybe to kick it off, where's the industry moving architecturally? Let's start with scale up first, and then we can dig into the scale out and scale across too.

Frank Chang: Okay, definitely. I think you hear a lot of topics regarding the scale up. So, first it's like from companies, like Nvidia and also that many startups, they're using corporates to dominate for the scale ups. I think for us, we do optical transceivers. We mostly focus on the scale out, because that is optic which dominates. And, also focus on the standard cloud networking. So, scale up definitely is one area right now, it's so hot, so many com startup companies, including some silicon photonic companies, they expect to yield in the optical to replace copper for the next three, five years. So I'm not sure that answers the questions you have.

Wamsi Mohan: Sure. Like, maybe just to dig into that a little bit. What do you think is the time frame for this transition to see optics as scale up, particularly as it pertains to Nvidia?

Frank Chang: I think for Nvidia currently, if using Jason Quong's term, they say we'll stay the copper as much as we can. And once the corporate starting to basically starting run out of gas, we move to optical. So that's the original words So basically, use copper as long as you can. Because, mainly the power consumption is lower, and also people believe copper is more reliable than optical. I think it's basically, I think Jason mentioned clearly, I think last year at the GTC conference, they're showing clearly, for example, optical still probably more expensive compared with copper. It basically for scale up. The

distance is normally like a very short distance, like one meter, two meters. So for that kind of distance, they stick with copper as much as they can. I think it [overlapping voices 06:37] talking about OFC, a lot of panelists talking about that. So, I think pretty much the general picture is copper probably stay pretty long time for the scale up.

Wamsi Mohan: Okay, so are we saying that the expectation would be that copper is there through maybe 2028 as the dominant technology, and then 2029 maybe we start to see adoption of CPO?

Frank Chang: Yes, I think of it this way, copper definitely will last probably through the 2028, but there's some other technology will penetration that. Probably you hear for the industry, they have the architecture called the slow and the wide. Slow white means for the GPU to GPU connection, because you don't need high speed You need the multiline, many multiline, like they just mentioned the slow and wide. You can connect it like, for example, if you want to create it 800G, you can use the 2 gig. You can use 40 of those. From copper to optical, if you hear also from Microsoft, they have the macro [indiscernible]. Macro [indiscernible] is one way to connect optically between the GPU to GPU. So that's that technology there. But [indiscernible 08:13] definitely right now is the bottleneck. So, people spend many technology there. So you probably, there's so many AI startups, probably somebody will adopt this sooner than 2028, but my understanding from the mainstream standpoint, 2028 will be the good timelines to probably you see a major customer adopt optical.

Wamsi Mohan: Okay. So I want to get to each of the hyperscalers and sort of what they are doing in a second, but before that, maybe we can just talk a little bit about the spend on fiber and transceivers within the data center and outside the data center. So maybe if we just start with inside the data center, can you give us some sense for maybe 100 megawatt data center, or roughly how many dollars are spent on fiber and for a gigawatt data center, how does that change?

Frank Chang: Okay, I can give some ideas. For example, basically it depends on the servers. So normally, for example, there's many companies studying, for example, basically the top tier hyperscalers, for example, even like Xai from Elon Musk. Even also Microsoft and also OpenAI, they're talking about the gigawatts. So starting with the 100 megawatts standard data center, especially for AI, that's probably normally, according to the formula from Elon Musk, normally they put 100k GPUs to connect them together So if you're using the latest the GB300, you probably reach beyond 100 megawatts. But I know that currently, for example, that Microsoft is starting to build like, I think it's in the fresh water data centers, they're starting to close the gigawatts. So, for 100 microwatts data centers, suppose you have 100k kind of GPUs, you can connect. It depends on how many GPU's to how many transceivers. Normally, for example, five transceivers or six transceivers, this ratio can be higher if you talking about 1.6T. So I know the 1.6T people talking about eight modules and nine modules, and even calculates how many modules you have and, for example, how many GPUs you have. Now you calculate the dollars associated with that.

For example, if you're talking about the transceivers, 800G modules normally, for example, around the 350 range right now, and the 1.6 modules still in the initial stage. But right now, the pricing drops very quickly, so right now it goes to 600 or 700 range. But eventually, because 800G module, 1.6 module have similar BOM [ph], eight receivers. And normally, the premium from 100 gigabit lasers to 200 gigabit lasers is kind of like 10%, 20%. So probably for the long term, you will see the 1.6T modules will be more cost effective. That's what people believe. So probably 800G will be doing more like saturated next year, and 1.6T probably last pretty long time. For example, we see 1.6T running up this year, should be able to probably dispense over to 2030, so before the 3.2T to pick up. So I'm not sure that answered your question.

Wamsi Mohan: That's helpful. So you gave us a ratio of transceivers to GPUs, but what about just sort of your fiber itself. So connectorized fiber, how many miles of fiber for

your example of like 100k GPU cluster, what is the right way to think about the density of fiber inside that data center?

Frank Chang: Okay, so for the fiber wide probably is different for each hyperscalers. Let me make one example. For like AWS or Meta, Meta normally use the two by four. When you do two by fours, normally you collect three kilometers. So three kilometers and the two fibers [indiscernible 13:29] means two fibers. So you can calculate if you have 100,000 GPUs, suppose you have let's put it module wise, so suppose you have 400,000 modules. And for the modules, that's pairs, that's 200,000 modules at module pairs, and each pair have like three kilometers ROC cable. So that's how you can calculate how much cables, fiber cables you need. You probably recently, I think probably last week, Meta just signed like \$6 billion with coding, so that probably is associated with this. But, as you know, currently, the single mode fiber for three kilometers is probably very cheap, but even that I think probably like few things per meter, but even that, \$6 billion, you can calculate how much fiber they want.

Wamsi Mohan: That's helpful. And then just in terms of outside the data center, what are you observing in terms of the spending as it pertains to scale across?

Frank Chang: Okay, so let's put it this way, before we talk about the scales across, the scale out is probably right now is the dominator version for all the optics usage right now. If you probably know, currently the 800G, rough estimates this year is like \$60 million or \$70 million. This is mainly used for what we call the scale out. So the scale across normally in the cloud wording, in cloud methodology, we call it the DCI. So, the DCI normally connect data centers to data centers, for example, like 10 kilometers over 2 kilometers. And also for the scale across, you have current. You have 80 kilometers, 120 kilometer, like the CR kind of type of applications. So, for the current technology normally used for the scale, for the scale across, also.

So, I think that is the area Optica dominates. Everybody is using optical transceivers there and also, for example, use the higher power hungry, the 400G module 800G module currently, probably 2028 or maybe 2027 we're starting to initially deploy the 1.6T module. If you're checking this year, probably OFC time frame, you will see many companies starting to live demo the 1.6T module. My study of OFC is starting to like industry multi vendor interoperability for the 1.6T one module. That's where the primarily use for the scale across.

Wamsi Mohan: Okay, that's helpful. So maybe let's just go into each hyperscaler. I'm sure a lot of our listeners are going to be interested in understanding how the various optical players are doing at each hyperscaler. So, what would be helpful, I guess, would be let's go hyperscaler by hyperscaler, and say what are they spending on in terms of pluggable optics, in terms of laser components, in terms of the multimode, single mode fibers? What are they using today? What is the direction of travel for that? What are they using for DSPs? So, if you can just start with maybe AWS, and think through sort of what the various optical players, what kind of positioning and share they have? Would that be a good way to tackle this?

Frank Chang: Definitely. For example, we can focus on the several key hyperscalers So, for example, starting with AWS, we work with all of them and we are suppliers for some of them. So for AWS, for example, let's talk about optical only. You probably know, AWS is normally they live in the industry, low cost. So for the optic side, they deploy very cute volume for 400G. If you check, mostly like vendors like [indiscernible 18:46], it comes from AWS for 400G. So, they're still shaping a lot of 400G, huge volume. And they are mainly like 400 years, like DR4, that type of optics is big volume. And also, they are starting to merge to the 800G this year. So, it backs, I think, probably 2025. So the next version they starting to deploy for 800G DR4, two by DR4 or two by SR4. So currently they are running volume this year. That's huge volumes. And they have more suppliers now. For example, we know in the established one used to be like AWS, for example, used to [indiscernible 19:41]. They recently got more lax and they recently got Cisco. And

also there's some startups, like Luminance. So these are their suppliers. They have lots of suppliers that they ask very low cost. Another optics they are using, they call the current light. For example, they say, I want in 20 kilometers, I want using the current to make the 20 kilometers, but I want to maintain the cost, \$500. \$500 for 20 kilometer current, that's I'm copying like rise is the developer, the product for them. So that's kind of working, for example, for AWS. Working for AWS, they have very special kind of a BOM requirements. For example, if you do the 800G you have to use ST at macro. If you check the news, AWS have collaborations with SG. So basically, you have the OCD for Silicon Photonics, you using SG photonic chip, and you have to use that method tool at the minimum requirement.

And then, most importantly, for AWS, different with everybody else, they merge to I3C. So for the management interface, everybody at Arrows is using I2C. They move to I3C. That is the requirement if you want to be their supplier. So that's AWS. Another thing we may talk with Google. For Google, recently because of the mini, I think the mini are showing better performance than OpenAI, especially talking about in China. The supplier become suppliers to Google become the very popular right now, because they believe Google the volume wave for next year, 2026 or 2027 we probably 3x or 4x increase. So, Google, basically, they are using AWS to buy. So initially, two by 400G. Right now is two by 800G. So two by 400G, this what they start in shaping the 1.6T module since the Q4 2025. So that volume, for example, if you compare with the 1.6T module shipment for 2025, pretty much dominated by Nvidia and Google. And Google definitely it look like it's in the uptrend compared with the Nvidia. All Google's architect based on what we call based on OCS, so they have the requirements using the optical switching. So that's Google.

And also, we can talk about Meta. So, we are the key suppliers for Meta. And Meta in 2026 will be our number one customer. Probably contributes a billion dollar kind of revenue for the company. So, I think for Meta, for 800G they are just starting to slip in volume last year. So this year, probably, Meta itself will contribute over 10 million units for 2026. So hopefully, we can get that 20% shares. That's 2 million at least, 2 million ports, at least. So Meta have slight different technology, different methodology to manage the suppliers compared with Google. But because, for example, Google, you know that it's using Cloud Light, using finisher, they have dedicated suppliers. So, it's very hard to get into the supplier for Google, because the yielding stables the suppliers all the time. Maybe this year, because of the shortage, they will starting to get false supplier, but that's Google's on how to manage [indiscernible 23:51]. But Meta has a similar methodology. They have sports suppliers, for example, they have [indiscernible 23:57], we are the first one we replace Intel because Intel got out of the transceiver business from 2023, so that's for Meta.

And another one we can talk with is Microsoft. We are also case suppliers for Microsoft, for example, the 400G, we are their single source for the 400G single mode application. But I didn't know, for 800G and the 1.6T, Microsoft is working with Nvidia closely, and Microsoft is probably the largest customer for Nvidia. They buy everything from Nvidia, including the ROC, including the even building data centers. You probably know, from Microsoft, they recently to have, I think it's the fresh water data centers, they are starting to deploy the GP300. So that is kind of Microsoft business model. And most importantly, Microsoft have many suppliers. For example, the AOC suppliers. Also, Microsoft is using a lot of, for example, 400TR compared with the other hyperscalers. And Microsoft, normally they call the disaggregation, basically they avoid to build a very huge data center. They build like more data centers around each other and each data centers is like 70 kilometers, 80 kilometer apart. So if you check Microsoft, they have between data centers connections, a 70 kilometer or 80 kilometer apart. So that is a slightly different architecture for Microsoft.

Maybe the next one we can talk about Nvidia. Nvidia, as we know, is completely AI, they starting with AI. So a GPU kind of connection for the AIS. They dominate the GPU

markets because others like Microsoft, like Google, like Amazon and Meta, they have the cloud. They have cloud data centers. But that is something different with Nvidia. Nvidia are doing what they call AI factory. So they build everything for AI. That's why it's also calling why their market cap is so high, close to \$5 trillion. So, for Nvidia, they are worse normally because GPU connection. They're normally a very short distance. If you compare with [indiscernible 26:58] their average distance is 150 meters. That's why, for 400G, 800G, they use a lot of multi mod. Multi mod normally covers 100 meters. So recently, because of the baud rate increase, they started into using DI8. So DI8 is moving to single mode, because the [indiscernible 27:21] doing the multi mods have problem to go beyond the 100 gigabit per seconds. For 200 [indiscernible] the DR8 cap application, normally building the either EML or using [indiscernible] becomes the dominant applications.

So that's why, if you see [indiscernible 27:43], they show you every GGC meeting, so they show their new architectures mostly for the scale out, but also the scale across. So using the, for example, the optics based on the 1.6T, even most of DI8's. I know they have very few probably demand for the two by 400G. Somebody mentioned the short distance probably is like 80% and probably 20% based on like two kilometers.

I'm not sure that covers most of the hyperscalers. I know others like Oracle, that's another one. And anyway, they are compared with this five is much smaller scale. And Oracle recently got a lot of attention because of OpenAI, Nvidia and then, because this kind of a circle financing. We call the financing, I don't know how to spell that, because Nvidia and OpenAI, Oracle this kind of financing circle.

Wamsi Mohan: Sure. Do you have a perspective also on the China cloud?

Frank Chang: Yes, of course. We deal with the China market like Ali, for example. We deal with Ali, with ByteDance and also Baidu. So, I think for the China market it's kind of two years, more than two to three years behind. For example, China still dominates by 400G. And the China situation that the entire most multi model, that this is normally like 100 meters, 200 meters, because the China is strange, the government regulations, anything beyond the two kilometer, you have to go to the carriers. You have to go to China Mobile, China Telecom, because they have the license if you do outside of the campus. But, for example, if you want to work with Alibaba, the distance normally is within one kilometers. And [indiscernible 30:06] probably have similar situation. We're not in the [indiscernible] supply side, but we definitely working very closely with Alibaba. And also, while we have some cloud, also. While we have some cloud, Ali cloud, and most importantly, for example, that's others like ByteDance, H3C, those kind of companies is working, for example also for the AI. But, mostly Chinese, because they're unique positions and also limited by the US kind of DSPs, the building materials, they're focused a lot about the being on the models So, for example, Tencent's and [indiscernible 31:04], they have their own AI model. And they have a lot of startups, like Deep 6 doing the AI models. So, it's competitive in the world.

Wamsi Mohan: That's helpful. Everyone on the call, if you have any questions, you can either hit me up on Bloomberg or you can raise hand and ask, or if you're dialed in, you can do that too. So let me just pause for a quick second to see if anyone on the line has questions. If you want to ask anything, go ahead and use the right hand feature or ping me, and I'm happy to ask on your behalf too.

Frank Chang: I have one comment for 2026. I think probably everybody hear there will be big material shortage. I can make one example. I think China banned rare earth to attending to Japan. So I think you can see that the latest supply from Japanese companies like Sumitomo, like Mitsubishi, I think will have a very big problem to supply like Enolite [ph], because they count on the leaders, because they don't have anything internally. So pretty much buy everything and then assemble together. So, I think for us, we have Taiwan fab, we have China fab. We don't have this limitation. So this gave us a very big opportunity to gain more market share.

Wamsi Mohan: Okay. Just to follow up on that, when do you see the laser shortages easing?

Frank Chang: I think laser shortages are already happening. If you listen from, for example, if you listen from some quantitative earnings from Lumentum, even from broad counts, I think they are talking about the EML's for 2026 would be short 30% or 40%. That's why you see people trying to using silicon photonics, because silicon photonics, you can one later drive four lines, you utilize lasers. That's the reason behind it, right? And people think the [indiscernible] share will be larger.

Wamsi Mohan: Okay. But when do you think that that constraint will become lesser? So, you're saying it's already getting constrained, but when do you think that that constraint will no longer be a problem?

Frank Chang: It's very hard for last year. I think it's very hard for these two years. If you think about it also even the DLTs, even DLT right now, the daytime used to be 26 weeks. Right now, everybody like Broadcom, in [indiscernible 34:16], they put like 45 weeks now. And many smaller companies, for example, startups, you want to build this, they normally put like what we call the minimum of MOQ, minimum of quantities. For finance, they even don't support many small startups. So that kind of situation, it will be very, very bad for 2026. Hopefully, because the lead time is long, if you want to have your supplies, normally you have to place your PO one year ahead. So I can think 2027 probably still tough.

Wamsi Mohan: Yes, okay. I have some questions here from the audience. One is asking if you can share any insights between UA Link versus NV link? And also, Ethernet versus PCIE for scale up switching, and what happens when we move to CPO?

Frank Chang: I think CPO is definitely a very big topic. If you're talking about like OMCs, you can see CPO, everybody talking about that. And from a research standpoint, CPO is a very good research project. But I just want to, from my observation, from the deploy standpoint, if you hear, for example, the Broadcom CEO, I think they mentioned recently, I think I mentioned from Broadcom's standpoint, Broadcom is doing CPOs. I think their original words are that the CPO won't matter in the next three, five years. That's what its original was. So basically, my understanding for the CPO standpoint, I should not say completely no. So currently, the five hyperscalers, only Meta say they may have interest for CPO. And if you check with the Jason for the SpectraX, also the QuantumX, they announced that probably will starting to doing production end of 2026, but they mentioned some like startup, like how waves they are waiting to starting to deploy some CPOs. But so far, to be honest with you, CPO probably is still not mainstream. Pluggable probably will be still dominate the markets with major market shares until 2030. So if you're talking about, for example, the market analysis from light counting, those kind industry, pretty famous kind of a marketing reports, the market share for CPO for 2030, beyond 2030, probably waiting like 5% or 10%, even less than 10% range, single digit, basically. So that's still kind of a plan for now. I know China used to talk about this for two to three years, for the last two three years. So far, I don't see anybody really waiting to starting to deploy CPO because of the risk, the flexibility, those kind of problems. Many of the problems, for example, Broadcom is doing CPO, Nvidia is doing CPO. Nvidia is a closed ecosystems, but Broadcom is trying to sell chips for enable the CPO market. But you still, if you do a CPO, you are dealing with very limited suppliers. So, you basically log into a few suppliers. So, for that kind of situation, it's different with the best [indiscernible 38:26] doing the pluggables.

So, my standing from the short term, maybe the RPO, NPO will probably see better shares compared with CPO. According to, for example, the industry like the speakers, for example, [indiscernible 38:52]. So, according to their wording, so the CPO can only become significant when the pluggable run auto manage. But currently, pluggable can run two, three pound 2T, no problem. So my standing at OFC right now is still secret, so there will be a new form factor called XPO. Put XPO basically is similar to the CPO idea,

but I'm not using, for example, one CPO peak with the switch together. But I will put like 6.4T and the 12.80 block modules, I can use like four of those modules and connect with the ASIC. We still can extend the pluggable to like 12.8T. So in that kind of situation, my understanding is the CPO can be pushed even further. So that is what observations we see from the industry.

Wamsi Mohan: Okay. And I'm not clear, unpluggables, how do you see the positioning of some of the traditional connector vendors like Amphenol T connectivity, Molex, and also maybe Corning, relative to other players?

Frank Chang: Oh yeah. I think for the higher baud rate, if we go to like beyond 3.2T, also going to like the 3.2T because you need a lot of higher speed connectors. This is the expertise following those connector companies like T Connectivity, Molex, that's their territory. So, you will see they gain a lot. And also, right for those kind of companies, they also running the deck cable, the AEC cables, so I think that's their territory. So, they will benefit from this kind of we call it the higher density connectors. Because the current industry, from the interim like OHF, even we're starting to talk about higher density connectors, because you want to make more advanced modules beyond the 1.6T. So there's so many discussions right now. My understanding, OFF we're starting to talking in February, and I think it's a possible [indiscernible 41:30]. So I think in mid-February, so that is one of the big topics.

Wamsi Mohan: Can you also talk a little bit about AEC, and where that fits in, in today's implementation, and what is their trajectory for that?

Frank Chang: Oh, yes, definitely. AEC is a huge market. I love it that way. If you can see, so companies like Credo, their market cap is just because the AEC, because they don't have much optics, that you can realize how important the AEC market is. The AEC market is basically for 1.6T because 1.6T, the DAC cable will run out of mileage, run out of gas, and also AEC probably people will not less use. I think that the AEC will become very, very important for the 1.6T. Because, for example, if I had five meters, seven meters, this typical distance, you have to use AEC. So that's why you see right from the 800G, you see ACC, you'll see DEC cables, and they also you see the AEC cable. But the one positive primary will be AEC cables, So AEC will be very important when we go to the 1.6T. But I'm not sure even for the 3.2T what distance will cover for the AEC, but the 1.6T you can still cover five meters and the seven meters. You have no choice for that.

Wamsi Mohan: There's another question that came up that's asking about Marvell Tech acquiring Celestial for scale up. Will Amazon move to optical scale up with Celestial?

Frank Chang: Yes. So I used to work for InPhi. I still have a lot of friends live there. So, to be honest with you, many people feel why buy the startup without clear products. To be honest, I've know them for a while. I don't know what they are doing, actually, but they bought like were significantly like, I think, three billion, potential to five billion, mainly because AWS is an endorsement So, but nobody has a clear picture what their product looks like. We never see their demos. If you compare with another startup, at the new builds, [indiscernible 44:02] bought the new builds, the new builds definitely have some production. They show the PCIE connection, the safety lines. We see that product. They demoed at the design con, and also they have the YFC type. But I don't know for Matt Murphy, from by way of Salistica Lab, so I don't know. So that part, even there are a lot of employees cannot understand that. But I think it's the only the projection that they are doing some kind of like design services or design projects with AWS. Potentially, they may have AWS business, and they claim they're working on scale up eventually they may get the AWS business and the growing business there. But to be honest with you, I think that's product still probably sometime away. My understanding is Matt Murphy had told all hand meeting in [indiscernible 45:20], you think about this is something like four years ago, they buy [indiscernible], but that's a slightly different situation. In fact, they have already have 700 kind of meaning kind of a revenue for 2022. So under this, I think [indiscernible] pay them like \$10 billion for that 700. Many

people think that time is too expensive, but you see they have clear revenues. So right now, I think Matt Murphy bought the suite, they spend three billion. I don't think they have any revenue so far. So anyway, it's hard to judge, but they definitely gave [indiscernible 46:09] definitely want to go to the scale up business. But scale up is true. It's a bottleneck right now, because all the bandwidth and everything. That is the hottest area. You will see probably several hundred other startups to chase the scale up business. I believe that will be bigger market in the future, but I'm not sure [indiscernible 46:37] position is worth that.

Wamsi Mohan: Okay. I think you mentioned it a little bit, but it sounds like to me, from what you said, that for multimode fiber inside data center, Amazon's going to be maybe the larger user. So, when do you see that transitioning? Or first of all, is that correct? And when do you see that transitioning to single mode? Is that sort of dependent on the speeds at which they're going to be running? And what about the other hyperscalers? Is everyone using sort of single mode or where do we stand on that?

Frank Chang: I think it could be precisely. I think Nvidia is the one to move to using a lot of multi mode, so they are starting transition to single mode. For example, if you're talking about 400G and 800G, they dominate with multi mode. For example, for 400G, probably 90% is multi mode. For the 800G, people believe maybe 60%, 70% is multi mode and the other 20% is single mode. Maybe there's also something like [indiscernible 47:50] applications. So only for 1.6T, they have no choice, because they are the first one starting 200 Lambda. And there's no such thing for the multi mode. Multi mode pretty much so far stop at the 100G Lambda. I know IEEE started new projects doing the 200G Lambda multi mode, but there's still no product yet. There are many demos like Broadcom doing demos, but still some gap to get for the full production, because the reliabilities and also become the distance support. So far right, the HV standpoint, so far the progress, even you can do the 200G Lambda multi mode, but the distance will be compromised, likely will 40 meters, 50 meters. [indiscernible 48:40] 100 meter, 120, 150 meters. But if you do 200G Lambda, so far my understanding is 30 meter, 50 meter is the maximum. Okay, that's kind of the situation for the expectation. So I think Nvidia, that means the one. For 1.6T, they moved to single mode, they moved to silicon photonics. So that's clearly. To be honest, with AWS, they are kind of not like aggressive like Google and Nvidia. So they are trying to maintain the low cost, so they are kind of delays, probably two, three year delay compared with others like Google, like Nvidia.

Wamsi Mohan: Okay, makes sense.

Frank Chang: [overlapping voices 49:37] ...two years away. Go ahead.

Wamsi Mohan: Okay, thank you. Operator, could we just give the instructions to people who are dialed in on the phone line as well, anyone who has any questions?

Operator: Of course. Again, if you wish to ask a question during the webinar, please use the raise hand button if you have joined via Zoom. If you have joined us on the phone, please press star one on your telephone keypad. We'll pause here briefly. We will now take the first question.

Participant: Hey, thank you so much for doing this. Credo was talking about this active LED cable, and I think you mentioned Microsoft might be using this. Can you elaborate on that?

Frank Chang: Yes.

Participant: How much time do we need for this?

Frank Chang: I can give you a quick answer. We work with Microsoft closely. I think the Microsoft R&D team, they call the slow and wide, they're starting to promote the macro LED kind of an idea, because that one fits very well, the GPU to GPU connections. You don't have to merge to basically using studies right to convert the slow lines, the slow lines to high speed lines. So Mac and LED definitely fit this application. That's one of the

reasons why Credo is starting to get into this business, because this will eat their AEC market share. Because normally GPU to GPU connection using AEC, so this is one of the technologies they tried to maintain that AEC kind of market share. That's why they're doing that direction. But to be honest with you, from my standpoint, with the optics expertise, we're waiting to go to the higher, we call it fast and narrow. Fast and narrow is probably the cost effective ways for the deployment. Because if you had slow and wide, typically you cannot avoid the linear, the linear of the cost, the kind of methodology. So, because you still have like, for example, if you want to use 40 liters to do like 800G, you still have so many kind of BOM's. Even your LED is cheaper, but it's still multiple LEDs. You cannot achieve what we call nine linear cost reduction.

Participant: Is it cheaper if you do it for like 40 meter? Maybe that's why it only works for 40 meters, 50 meters? Do you think up to 30 meters or 40 meters or something?

Frank Chang: It's cheaper from each module standpoint, but it's still your lasers is too many that way, so I don't know how they handle fibers. I think it's not the way from optics standpoint. So, I don't want that. I think for the market wise, we may look and see, but historically, people waiting theory, wake up this way. Serial always wins than parallel. For example, previously we do like 4x100G for 400G and that will goes to 4x20G, goes to 800G. So, this way is normally cheaper. To make one example, your 10 year later, if you go to 25 years, you probably increase 10% of the cost—

Participant: But if you start using a laser, it's using an LED.

Frank Chang: I know, I would do the same thing. You still have 40 LEDs, right? But later, right now, already cheap. Even the LED is very cheap, 40 of those. Anyway, I don't want to [overlapping voices 53:43] low speed thing. I know it's a countdown allowed on the city cap 20 integration. But so far, the market has not made any decision yet, but definitely they have one way to deal with the GPU to GPU connections. They have that position for sure. For example, GPU starting with the low speed, multi channels, low speed, so they are fit that structure very well.

Participant: Got it. Thank you so much. This is great. Very helpful today. Thank you.

Frank Chang: Yes, definitely a hot topic. So, I think probably in the year, because technology wise it's not difficult, so probably in a year or two years, we will see the market's decision.

Wamsi Mohan: Okay, great. Thank you for that. I think maybe we have time for one last question, if there are any questions on the line.

Operator: Yes, we may take the next question.

Participant: I think you kind of sounded bullish on AEC, which is mainly Credo's market. If we move to CPU, AEC will become more important or less important and why? Thank you.

Frank Chang: Let me put it this way, currently, the AEC market, AEC normally is like 5 meters to 7, that's played very well for the GPU to GPU connection, we call the scale up. But my understanding, if I read correctly from Broadcom, from Nvidia, the CPU is more focused on what we call on the scale out. So if you go to the scale up area, you want to be using, for example, more like people try to talk different ways. Like NP, NPO, those more flexible kind of connections. So if you're talking, for example, where they dominate the CPU markets, most people think that they scale out. It's not scale up. So I'm sure I answered your question.

Participant: No, I think you kind of hinted at it. Yes. I think the way I'm thinking about it is as you move on to CPU, or maybe potentially 3.2 CPU as speed goes up, I feel like you're probably right. AEC becomes far more important than just direct attached copper, because direct attached copper, obviously, you have signal integrity issue versus AEC you don't have that issue.

Frank Chang: Yes, when we talk with our customers, especially for 1.6T, I see customers more interested on the AEC side than to give you real POs, they move ahead with, for example, the call everything. So, I think for others, especially for 1.6T for this year, I see others is still kind of hesitation, and we still receive RFR's regularly from our customers. But so far, I think it's still kind of in the research stage. And for the CPU standpoint, if you're talking about the five hyperscalers, my take is that only Meta have some kind of a [indiscernible 57:27] they say, I'm still sensitive to the power. I will look into the LPO. I will look into the CPU. But others, for example, Google they know, AWS they know. Microsoft used to yes that they know. Even , they have R&D people talking about that, they're production people say no. So that kind of situation is left that way. And Nvidia has, for example, the Quantum X, the Spectrum X that mainly is, for example, starting for it. You get replacing the 100P switch right now. So hopefully they can manage to manage the manufacturer mass for the CPU, so they can manage to shaping like some kind of volume. But I don't see their major customers trying to do the CPU. My understanding, for most hyperscalers, they have systems. They know how to qualify the unpluggables. They have whole internal systems, the core systems to qualify that. Mine is standing, nobody really know how to qualify the CPO right now. Probably you need a complete new system. You establish a complete new systems to qualify the CPOs. Even pluggables, you spend normally six months or nine months to qualify one supplier. I don't know how to for CPO side if you have problems, how to debug that. If you have yield, how do you handle that. Those kind of things definitely is not flexible like pluggables.

So, we will see. We'll get better picture probably after 2027, how the CPO will manufacture.

Wamsi Mohan: Okay, great. I think we're just about out of time. So, I appreciate everyone joining. Frank, thank you so much for taking the time, and we look forward to chatting with you again.

Frank Chang: Yes. Thank you very much. It's very helpful. I think it's a lot of very good topics. I know I just saw the emails. We did not cover all the topics, but we can definitely talk next time. I know [overlapping voices 60:07] a lot of innovations. But I encourage everybody to attend the OFC. It covers all the latest innovation for leaders, DSPs, pluggables and fibers. I already mentioned, FPO will be like a big, big news for OFC. Because right now, I might say, try to cable to keep a secret for everybody, they want to make like a big news for the OFC. That will be the next pluggable future to tackle the CPU.

Wamsi Mohan: Who do you think is the biggest beneficiary of that XPO announcement?

Frank Chang: I think [indiscernible 60:50] company will benefit, for sure, because it's completely new mechanical housing, and also the new high density connectors, and you can allow like 64 channels. So, for example, if you stick with 200G plan, you can do like 64 of those. Basically you can get 12.8T. So that will give you the picture. For pluggable, it goes to beyond the 3.2T. You can go directly to 12.8T. So that's the basic message. And so far, the module will directly using what we call the liquid cooling. They will have liquid cap, so the water will running through the module. So that's kind of a structure is very similar to the CPU idea, but using pluggable approach. You can find that. It will be interesting. This OFC will be interesting and is just probably five weeks away.

Wamsi Mohan: Amazing. All right. Well, thank you so much. We really appreciate you taking the time, Frank. Thank you.

Frank Chang: Thank you very much. Thank you, everyone.

Operator: Thank you. This concludes today's webinar. You may now disconnect from the call. Thank you.



Abbreviations

AEC: Active Electrical Cable

AI: Artificial Intelligence

AOC: Active Optical Cable

ASIC: Application-Specific Integrated Circuit

ASP: Average Selling Price

AWS: Amazon Web Services

BOM: Bill of Materials

CPO: Co-Packaged Optics

CR: Coherent Reach / longer-reach optical link category

Comm: Communications

CTO: Chief Technology Officer

DAC: Direct Attach Copper

DC: Data Center

DCI: Data Center Interconnect

DR4: 4-lane Dense Reach optical module

DSP: Digital Signal Processor / Digital Signal Processing

EML: Electro-Absorption Modulated Laser

GTC: GPU Technology Conference

GPU: Graphics Processing Unit

I2C: Inter-Integrated Circuit

I3C: Improved Inter-Integrated Circuit

IEEE: Institute of Electrical and Electronics Engineers

LPO: Linear Pluggable Optics

MOQ: Minimum Order Quantity

NVDA: NVIDIA Corporation (ticker)

NVLink: NVIDIA's high-speed interconnect technology

NPO: Near-Packaged Optics

OFC: Optical Fiber Communication Conference

Oppty: Opportunity

OCS: Optical Circuit Switch

PCIe: Peripheral Component Interconnect Express

PO: Purchase Order

R&D: Research and Development

RFR: Request for Review

RPO: Re-timed Pluggable Optics

SR4: 4-lane Short Reach optical module

T: Terabit / Terabits per second

Tech: Technology

USD: United States Dollar

US: United States

XPO: Next-generation pluggable optical form factor

400G: 400 Gigabits per second

800G: 800 Gigabits per second

1.6T: 1.6 Terabits per second

3.2T: 3.2 Terabits per second

MW: Megawatt

m: meter

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